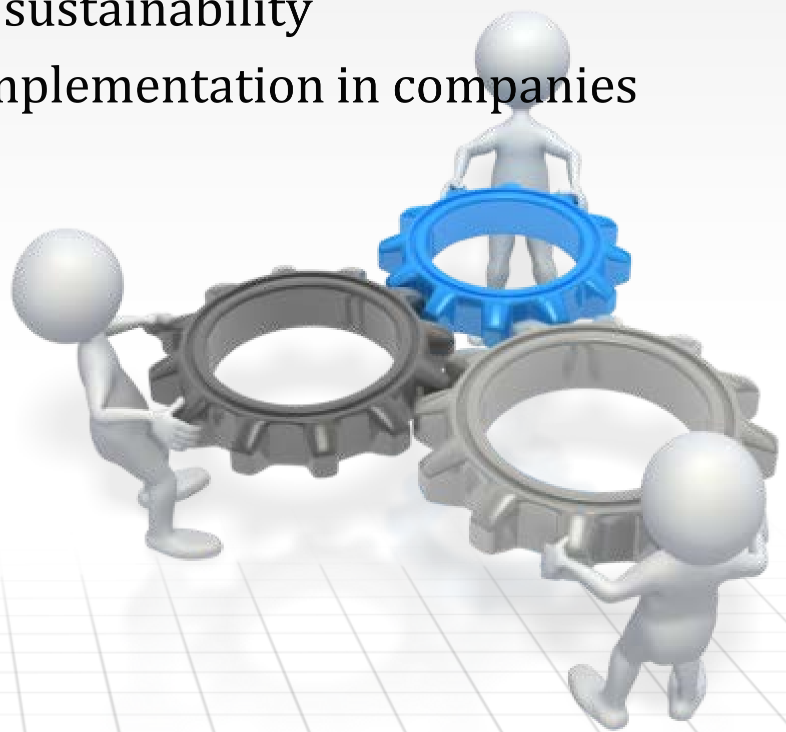
LECTURE 11

SUSTAINABLE MANUFACTURING

Contents



Objectives

1. Knowing the principles of sustainability.
2. Understanding the specific requirements of sustainable manufacturing.
3. Knowing the sustainability assessment methods in a company.
4. Knowing the application examples of the principles of sustainability in the companies in different fields.

Introduction (1)

***Sustainable development** - is the development that meets the needs of the present without compromising the ability of future generations, to meet their needs.*

***Sustainable manufacturing** - ensures the manufacturing of products using processes that minimize negative environmental impacts, conserve energy and natural resources, are safe for employees, communities and consumers and are economical.*

Introduction (2)

Sustainability is a journey, not a destination.

- There are no companies who by their activity have no impact on the environment.
- The main objective of any company should be to continuously implement sustainability at all levels.

Environment and production

- In sustainable manufacturing the focus is on:
  - making products using fewer materials and energy,
  - reducing waste in the manufacturing process,
  - avoid using risk materials,
  - development of "green products" (recyclable, a small amount of power used for their production etc.).
- Clean technologies, also called green technologies are technologies that ensure:
  - environmental protection,
  - compliance with environmental legislation,
  - pollution prevention and control,
  - waste management,
  - development of environmentally friendly infrastructure.

Examples of clean technologies: solid waste management, wastewater treatment, recycling, the use of solar panels, the use of wind turbines.

Self-assessment test

1. Sustainable development compromises ensuring resources for the next generation. **T** **F**

2. Sustainability assurance is a destination for any company. **T** **F**

3. In sustainable manufacturing focus is on making products using fewer materials and energy. **T** **F**

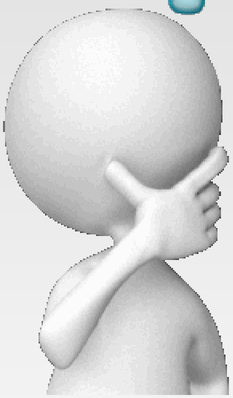
4. The sustainable manufacturing involves the removal of waste from the manufacturing process. **T** **F**

5. Green technologies provide pollution prevention and control. **T** **F**

Green products

A green product is a product designed to reduce its environmental impact.

- It is very important to consider the environmental impact of a product from design phase.
- Methods for manufacturing green products:
  - use of renewable resources,
  - reducing the number of materials for a product,
  - increasing the efficiency of production processes,
  - minimizing the use of packaging,
  - integrated development of products,
  - extending the life cycle of a product,
  - development of recycled products.



Implementation of sustainable manufacturing

- It is quite difficult to meet the specific criteria of sustainable manufacturing.

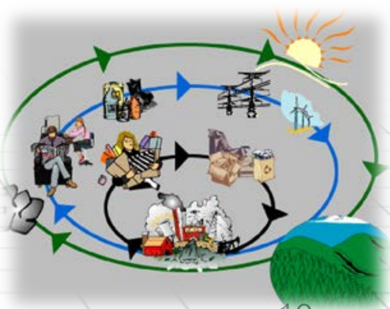
- Actions to achieve the objectives required by sustainable manufacturing:
  - improving practices used in manufacturing and maintenance,
  - process optimization,
  - replacement of raw materials,
  - using new technologies,
  - design for sustainability.

The industrial ecology

- In the past, companies focused on reducing pollution (the use of technology to reduce emissions).

- Today companies need to focus their efforts and resources on preventing pollution, to meet environmental standards and comply with the principles of sustainable manufacturing.

- Industrial ecology examines the evolution of materials and energy throughout production cycles and identify inefficiencies and losses in manufacturing processes.



Self-assessment test

1. A green product eliminates its environmental impact. **T** **F**

2. Environmental impact of a product should be considered from the design. **T** **F**

3. The products that minimize the use of packaging can be considered green. **T** **F**

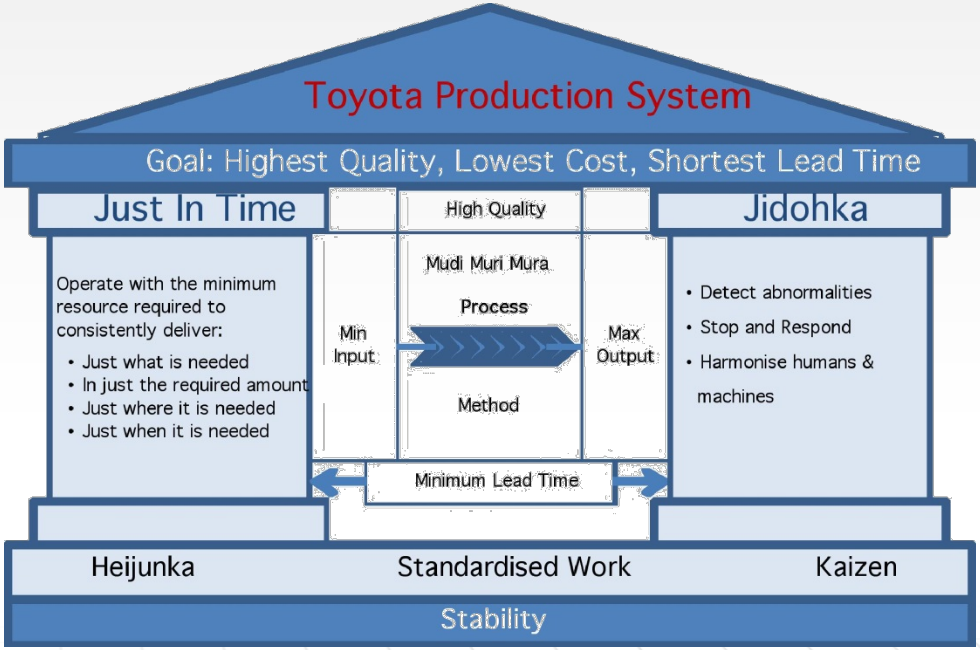
4. Optimization of production is a measure required to ensure sustainability in manufacturing. **T** **F**

5. Sustainable manufacturing is based on reducing pollution. **T** **F**

Lean manufacturing

- ❑ Lean manufacturing is a management philosophy derived from the Toyota production system.

- ❑ Lean Manufacturing removes:
  - defects,
  - overproduction,
  - delays,
  - inefficient allocation of workers,
  - transport,
  - inventories,
  - movement,
  - additional processing.



Clean manufacturing

- Clean manufacturing implements methods of environmental protection beside Lean manufacturing specific concepts.

- Clean Manufacturing provides:
  - more efficient use of materials,
  - responsible consumption of energy,
  - reduction or elimination of toxic materials from production processes
  - waste reduction:
    - solid and/or hazardous,
    - resulting from packaging materials,
    - released into the atmosphere,
    - discharged into the water.

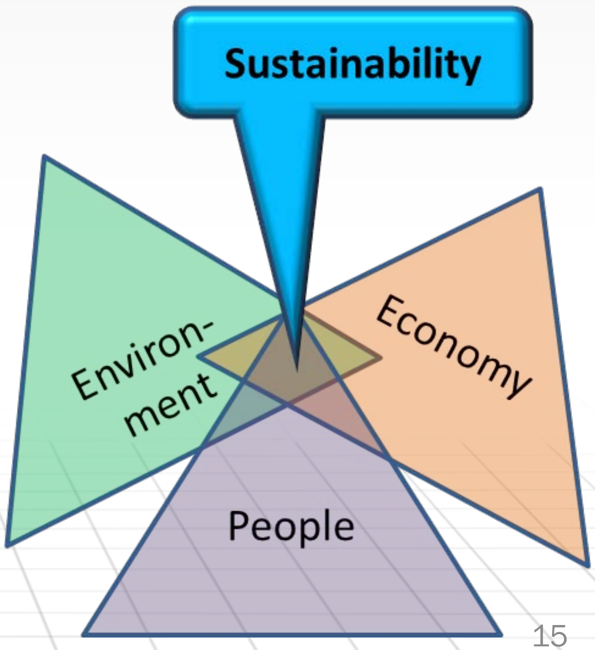
- Means of implementation of clean manufacturing:
  - maintenance measures,
  - substitution of raw materials,
  - effective control of processes,
  - improvement of equipment,
  - changing technologies,
  - regeneration and local reuse,
  - the manufacture of useful secondary products,
  - modifying products.

Lean and Clean manufacturing

- **Objectives of Lean and Clean manufacturing**:
  - **eliminating or reducing activities** that do not add value to a product,
  - **eliminating or reducing environmental impact,**
  - **identifying conflicts** between customer specifications and principles of a clean environment, and removing them.

- **Benefits of Lean and Clean manufacturing**:
  - efficient use of labor, time and capital
  - efficient use of raw materials and energy,
  - achieving maximum productivity using fewer resources,
  - creating a way to obtain an increase in production without an equivalent increase in the consumption of resources,
  - support the viability of a business.

- **Sustainability requires a balance between profitability, people and the environment.**
- Areas where the principles of sustainability must be implemented:
    - equipment
    - operations
    - cleaning,
    - storage and transport,
    - facilities,
    - resources certified by external institutions
    - culture



Sustainability through equipment

- Sustainability through equipment involves several steps:
  - looking for equipment whose design and implementation follows the principles of sustainability,
  - encourage providers to describe their equipment aspects of sustainability,
  - reducing production resources needed by combining multiple functions into one device,
  - improving existing equipment,
  - use of commissioned equipment,
  - marking using laser technology instead of ink printing,
  - donation instead of throwing.

Self-assessment test

1. In lean manufacturing there is no overproduction. **T** **F**

2. Clean manufacturing ensures responsible use of energy. **T** **F**

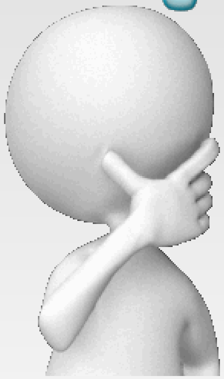
3. Substitution of raw materials with more polluting ones is a method of implementing clean manufacturing. **T** **F**

4. Lean and Clean manufacturing provides support for the viability of a business. **T** **F**

5. Donating equipment is not a practice encouraged by the principles of sustainability. **T** **F**

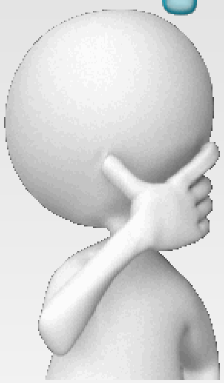
Sustainability through operations

- Sustainability through operations involves several steps:
  - increased use of equipment,
  - standardization to reduce the stock of material.
  - finding an answer to the following questions:
    - Materials and components are available when needed or they have to be ordered?
    - The operator can perform adjustments without using specialized tools?
    - Adjustments are made automatically or manually?



Sustainability by cleaning

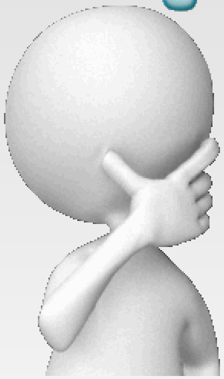
- Sustainability in manufacturing operations can be improved by using some alternative methods for cleaning up:
  - reducing water consumption,
  - reducing or eliminating the need for cleaning,



Sustainability by storage and transport

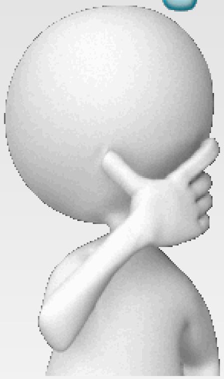
- ❑ Storage and transport activities must be optimized to meet the requirements of sustainability. It involves:

- reducing storage space,
- reducing energy consumption,
- reducing solid waste,
- using an integrated management system.



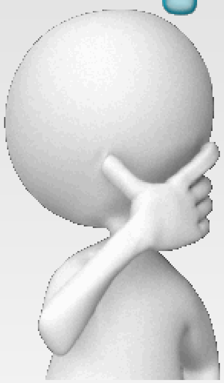
Sustainability by utilities

- ❑ Measures necessary to guarantee efficient use of resources:
  - eliminating air leakage,
  - reducing consumption when the equipment are idle,
  - reducing electricity consumptions for equipment,
  - reduce energy consumption for lighting.



Sustainability by using certified resources

- There are independent organizations that can help producers to implement those technologies that meet environmental requirements and more.
- Examples:
  - Green Seal Organization
    http://www.greenseal.org/
  - Green Guard Organization
    http://www.greenguard.org/
  - Forest Stewardship Council
    www.fsc.org/



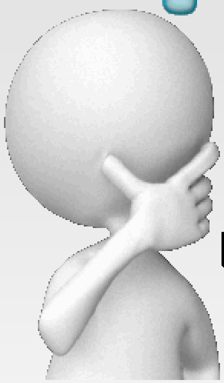
Promoting sustainability in companies

- Sustainability principles should be implemented in a company at all levels: from the manager up to worker.
- There should be a series of measures notified to all staff, which then should be implemented.
- Information shall be provided regarding the results of the implementation of sustainability.
- To motivate staff in achieving the overall objectives, it should be rewarded.

Sustainability assessment (1)

- Organization for Economic Cooperation and Development (OECD) * of which Romania is part, has developed a guide to help companies to implement sustainability principles.
- OECD has defined a set of 18 quantitative indicators as follows:
  - 3 for the materials used as raw materials,
  - 8 for production processes,
  - 7 for the manufactured products

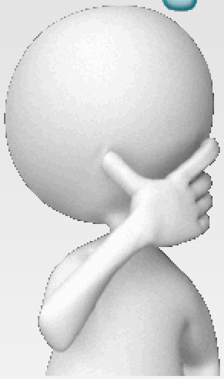
*http://www.oecd.org



Sustainability assessment (2)

- For raw materials (inputs into the production process) OECD has defined the following indicators:

- the level of usage of materials which are not renewable 
- the level of usage of restricted substances, 
- the level of usage of reusable materials / recyclable materials. 



Sustainability assessment (3)

- The benefits of environmentally friendly practices:

Lowering the amount of raw material	Reducing transport
Minimizing the amount of toxic materials	Reduce costs of handling, storage and treatment
Replacing toxic materials with some less toxic	Reducing the effort to respect environmental norms
Increasing the use of renewable and recyclable materials	Reducing waste disposal requirements

Sustainability assessment (4)

- OECD has defined eight indicators for analyzing manufacturing operations:
  - water consumption per unit of product,
  - energy consumed per unit of product,
  - proportion of renewable energy in total energy consumption,
  - amount of greenhouse emissions per unit of product,
  - the amount of waste per unit of product produced,
  - amount of emissions into the atmosphere per unit of product,
  - the amount of residue in water issued per unit of product,
  - percentage of the total area of the company, which is covered with natural materials (grass, trees, etc..)

Indicators 3 and 8 must have high values, and the rest as small as possible.

Sustainability assessment (5)

- Benefits of achieving environmentally friendly processing operations:

Reduction of energy consumption	Reduce costs associated with energy.
Reduction of waste and emissions	Reducing the cost of monitoring, treating and disposing.
Reprocessing of recovered materials at the end of life of a product	Increasing the added value of raw materials
Replacing or upgrading equipment / production lines	Improve efficiency of operations
Improving local biodiversity and natural conditions	Reduce the need for water, air conditioning and other utilities

Sustainability assessment (6)

- OECD has defined seven indicators for assessing the products:
  - proportion of a product that is recycled or reused,
  - proportion of a product represented by recyclable materials
  - proportion of a product represented by renewable materials,
  - level of utilization of non-renewable materials,
  - the amount of restricted substances in the product composition,
  - energy consumption of the product,
  - amount of greenhouse emissions of the product.

Indicators 1, 2 and 3 must have high values, and the rest as small as possible.

Sustainability assessment (7)

- The benefits of green products:

Replacing non-renewable materials with renewable or recycled ones	Reduce material costs. The development of attractive products.
Reduction of hazardous substances in products	Reducing the cost of monitoring, treating and disposing.
Improving the recyclability or biodegradability of products	Increasing the value of raw materials. Reducing waste costs.
Reducing the amount of energy necessary for the operation of a product	Decreasing the cost . Increasing the attractiveness of the products.
Increasing the products durability	Reducing the need for non-renewable materials.

Sustainability assessment (8)

- For the use of indicators, OECD recommends a sequence of seven steps, grouped into three stages:
  - Stage 1-Preparation:
    - Step 1: Impact analysis and prioritization,
    - Step 2: Select useful performance indicators.
  - Stage 2 Assessment:
    - Step 3: Measure the inputs used in production,
    - Step 4: Analysis of the production operations,
    - Step 5: Evaluation of products.
  - Step 3: Optimization:
    - Step 6: Understanding results of the analyzes made,
    - Step 7: Act to increase performance.

Self-assessment test

1. Implementing the principles of sustainability rests solely to the management team of a company. **T** **F**

2. Utilization of reusable materials / recycled materials must be increased. **T** **F**

3. The energy consumed per unit of product, according to the principles of sustainability can be increased if it comes from renewable sources. **T** **F**

4. Utilization of non-renewable materials must increase to reduce waste. **T** **F**

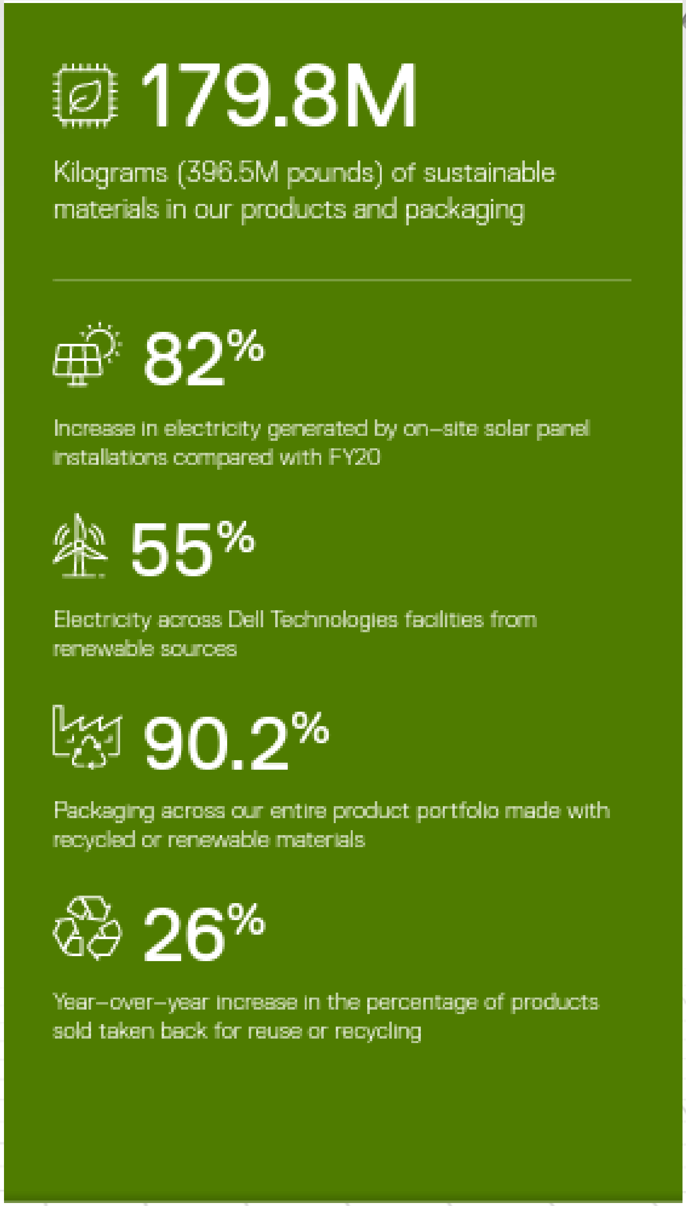
5. A product must contain as many recycled or reusable components. **T** **F**

Examples of application of sustainability in the companies (1)

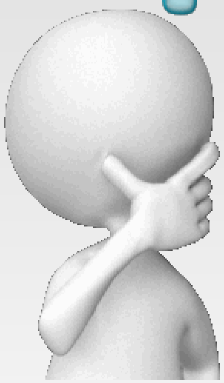
- Apple Computer publishes an annual report on the compliance with environmental restrictions and carbon footprint, available at http://www.apple.com/environment/ .

Examples of application of sustainability in the companies (2)

- Dell publishes an annual report on the compliance with environmental restrictions and carbon footprint, available at https://www.dell.com/en-us/dt/corporate/social-impact/esg-resources/reports/fy22-esg-report.htm

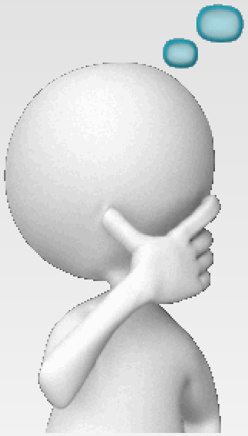


Examples of application of sustainability in the companies (3)



- The environmental report for 2021, published by HP, can be found at the address: https://www8.hp.com/h20195/v2/GetPDF.aspx/c08228880.pdf

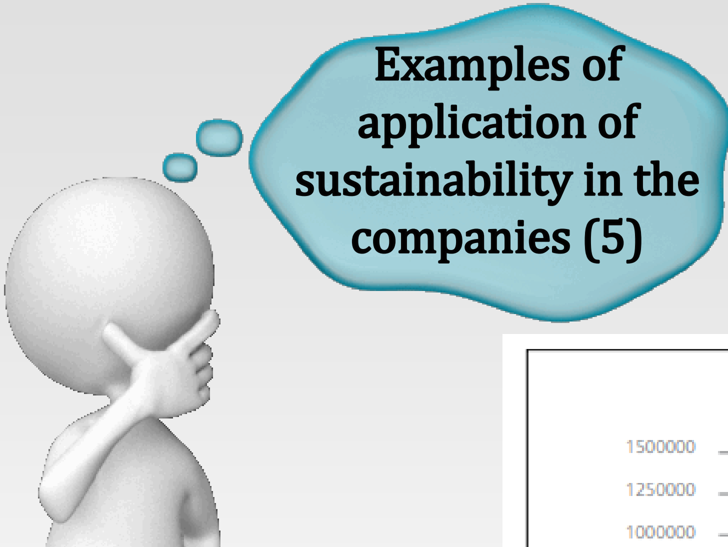
Goal	Progress in 2021
Carbon emissions	
Reduce HP value chain GHG emissions by 50% by 2030 (compared to 2019), and achieve net zero emissions by 2040[4]	HP's carbon footprint of 28,459,500 tonnes of CO_2e in 2021 was 9% less than in 2019, primarily due to reductions related to product use resulting from increased energy efficiency and changes to the mix of products sold. Learn more.
Reduce Scope 1 and Scope 2 GHG emissions from global operations by 60% by 2025, compared to 2015[5]	HP's global operations produced 159,500 tonnes of Scope 1 and Scope 2 CO_2e emissions, 59% less than our 2015 baseline. Learn more.
Use 100% renewable electricity in our operations by 2025	HP's global operations procured and generated 264,054 MWh of renewable electricity and attributes, equivalent to 54% of our global electricity consumption. Learn more.
Reduce HP product-use GHG emissions intensity by 30% by 2025, compared to 2015[6]	HP has achieved this goal for the second year in a row, with a 39% decrease through 2021, compared to 2015 (therefore, we will not report on this goal moving forward). Learn more.
Circularity	
Reach 75% circularity for products and packaging, by 2030[7]	39% circular by weight.[8] Learn more.
Recycle 1.2 million tonnes of hardware and supplies by 2025, since the beginning of 2016	Recycled 764,800 tonnes. Learn more.
Use 30% postconsumer recycled content plastic across HP's personal systems and print product portfolio by 2025[9]	13% achieved. Learn more.
Eliminate 75% of single-use plastic packaging by 2025, compared to 2018[10]	44% reduction, from an average of 221 grams/unit in 2018 to 124 grams/unit in 2021. Learn more.
Reach zero waste in HP operations by 2025[11]	In 2021, we achieved an 86.4% landfill diversion rate globally. Learn more.
Forests	
Counteract deforestation for non-HP paper used in our products and print services by 2030.[12] Continue to source only sustainable fiber for all HP brand paper and paper-based packaging for home and office printers and supplies, PCs, and displays.[13]	During the year, we addressed 23% of our total fiber footprint for paper used in our products and print services. Our programs counteracted deforestation for non-HP paper representing 19% of this footprint.[14] Since 2020, all HP brand paper and paper-based packaging for home and office printers and supplies, PCs, and displays have been derived from recycled or certified sources.[15] During 2021, these equaled 4% of this footprint. Learn more.
Water	
Reduce potable water withdrawal in global operations by 35% by 2025, compared to 2015, focusing on high-risk sites	HP withdrew 2,245,000 cubic meters of potable water across global operations in 2021, 30% less than in 2015, and focused reduction efforts on high-risk sites. Learn more.



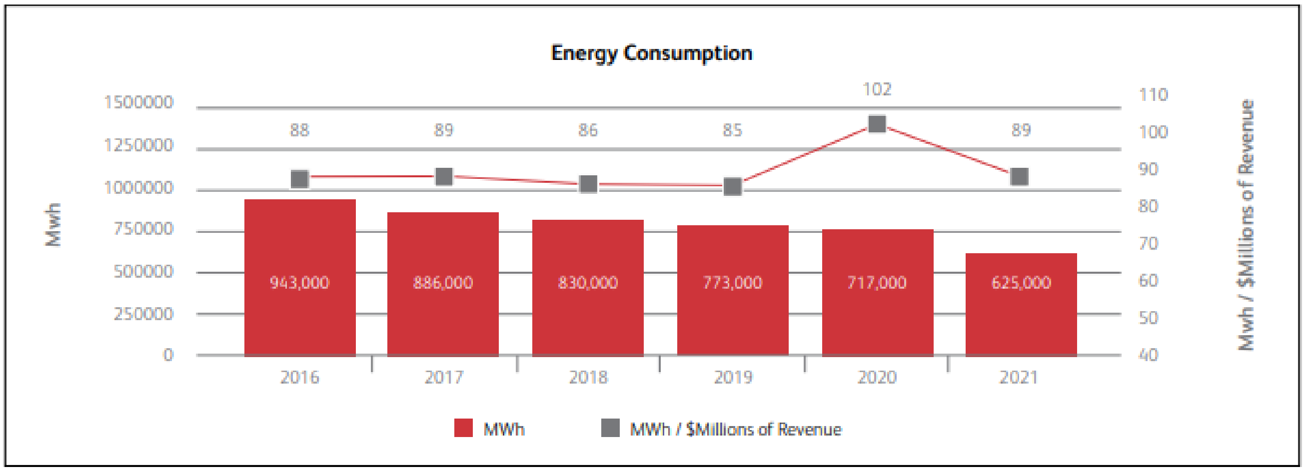
Examples of application of sustainability in the companies (4)

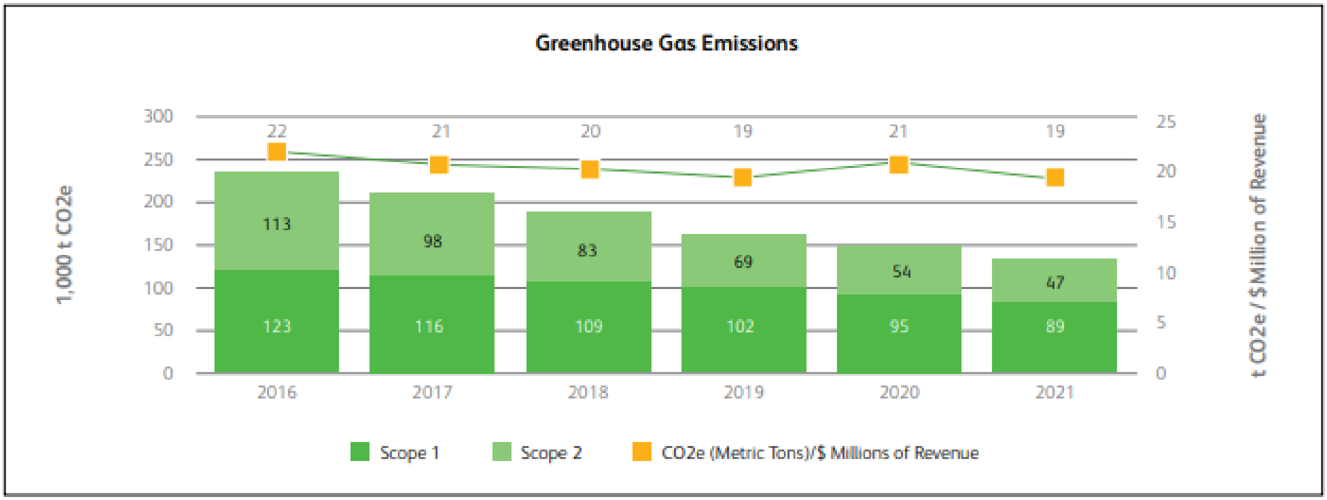
- The sustainability and financial report published by Ford for the 2023 can be found at: https://corporate.ford.com/social-impact/sustainability.html

Examples of application of sustainability in the companies (5)



- The 2022 report of the XEROX Company is available at: https://www.xerox.com/downloads/usa/en/x/Xerox_CSR_Report.pdf





Conclusion

- ❑ Environmental regulations are becoming more stringent.
- ❑ Sustainability involves cooperation between companies, local authorities, Government and European authorities.
- ❑ All efforts should be focused on ensuring sustainable development, satisfying current needs without compromising those of future generations.
- ❑ Despite the involved initial costs, on medium and long term the companies will benefit from the implementation of sustainability principles at all levels.

- ❑ Revolutions in manufacturing
- ❑ Cloud computing
- ❑ Big data analysis
- ❑ Internet of things
- ❑ Industrial internet of things
- ❑ Cyber-physical systems
- ❑ Digital factory
- ❑ Smart factory
- ❑ Industry 4.0 and 5.0